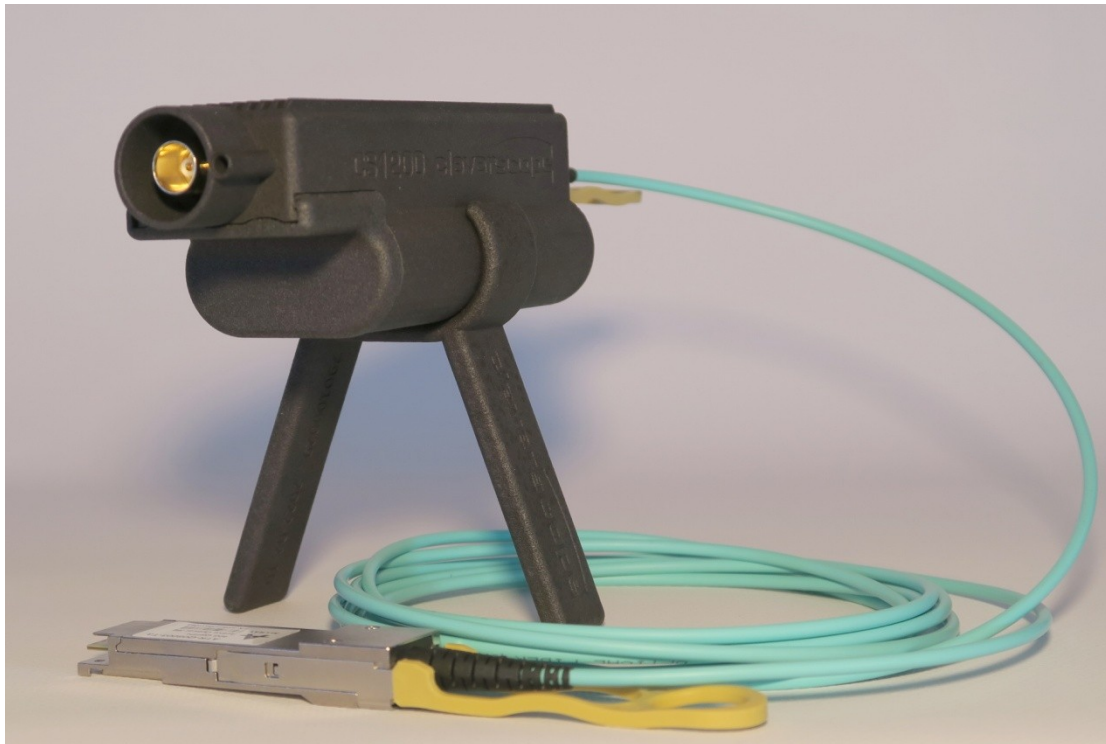


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CS1200 IsoPod – User Guide

Introduction

The CS1200 **IsoPod** is a remote isolated digitizer Pod for the CS548 which allows measurement at higher isolation voltage than the CS548 internal limit of 1kVAC CAT III or 2kVDC. It uses a standard QSFP terminated active optical cable (AOC) to connect the CS1200 IsoPod to the QSFP remote sockets on the CS548 front Panel. The input specification is identical to the CS548 internal channels.



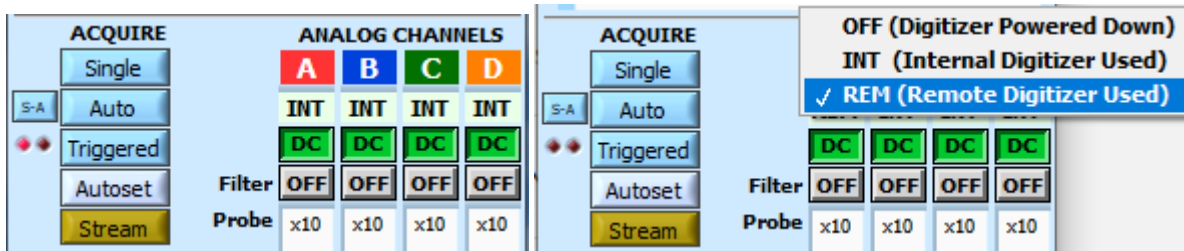
CS1200 IsoPod

- 30kV isolation provided >150mm spacing between IsoPod and other structures.
- 2 pF free space capacitance >50 mm above reference plane
- 100 dB CMRR at 50 MHz
- 14 bit resolution, 100dB dynamic range
- 200 MHz Analog BW
- 100uV resolution on 0.8v range
- 200uV rms noise on 0.8V range
- Two input ranges: +/-8V and +/-0.8V.

This user guide shows how to connect and use the IsoPod. Please refer to the CS548 manual for full specifications.

Application

The default setting is 'INT' for all channels. Click on the INT button to select REM for the remote IsoPod:



Once you have set a channel REMote, it will try and connect to a powered up CS1200 IsoPod, once available.

Batteries

The IsoPod uses 2x '21700' size Li-Ion cells. These are normally delivered in a semi discharged state so please fully charge before first use.

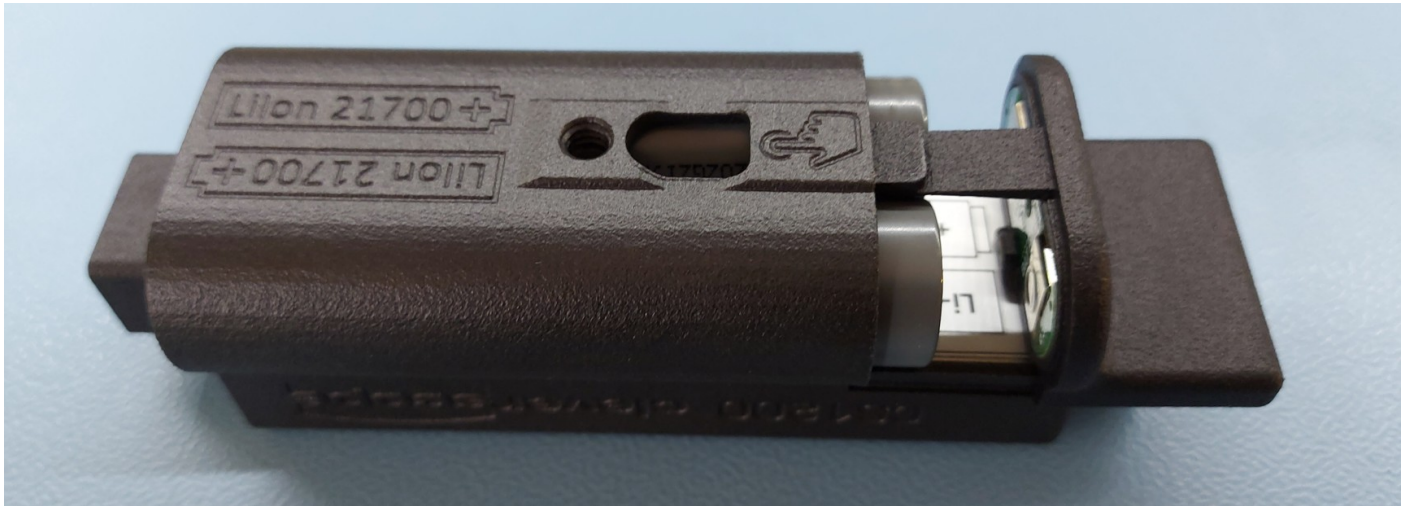
Note: The demo system is supplied with 4x 21700 5000mAh cells and 8-cell Li-Ion charger with USB type C adaptor.



Once charged insert 2 cells into IsoPod paying attention to the polarity as shown on the enclosure:



Then slide battery cover into place:



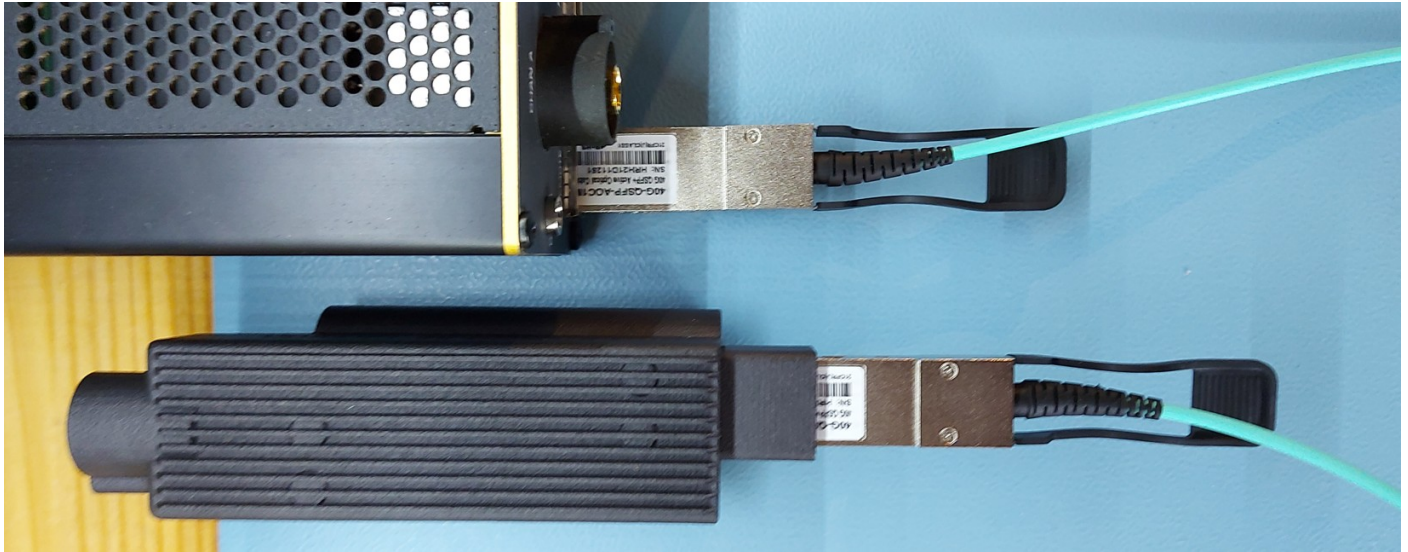
To open the battery cover, press the button and pull on cover:



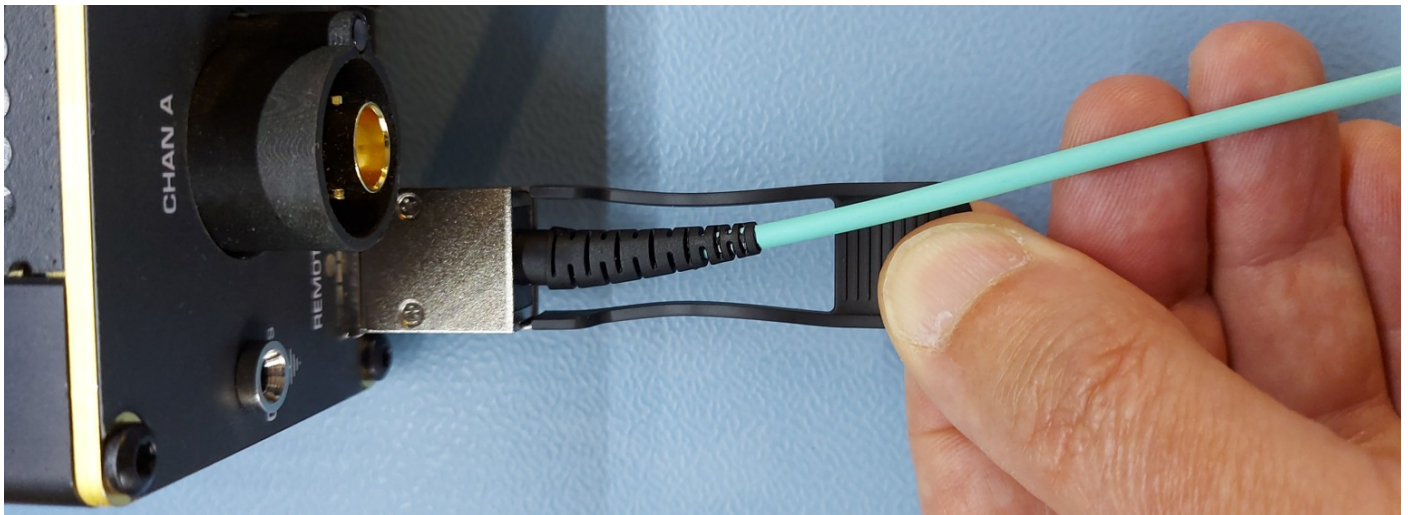
The supplied 5000mAh cells will run the IsoPod for approx 8.5 hours. At present we have not implemented battery monitoring in the application so keep in mind the amount of time the IsoPod has been running. If the AOC is removed the IsoPod will go into a low power state extending battery life. However if the IsoPod will not be used for some time it is best to remove the cells.

Fibre Optic Cable

The demo is supplied with a 3m long QSFP Active Optical Cable (AOC) but other lengths are available (2, 5, 10, 20, 30 & 50m). Insert the AOC into the CS548 and IsoPod with the label facing upwards:



Plug in the AOC metal QFSP with the pull tab under the fibre; push on either side of the pull tab close to the QSFP body until it clicks. It may help to use a screw driver tip against the metal body. You may need to find the socket with a bit of fiddling. To release pull on the plastic tab. **DO NOT PUSH OR PULL ON THE FIBRE, OR THE STRAIN RELIEF.**



Turn on the CS548. Once the power up cycle is complete, and the application has started (with the pod set to REMote), and connected (click on the Connect button next to Connection Status: **Connection Status: **, which should change to the **Connection Status: **) the IsoPod LED and enabled channel LEDs will cycle from blue (IsoPod found) to green (all channels synchronized):



Probing

The IsoPod is supplied with a stand placing it 65mm above the work surface. This will reduce the probe free space capacitance and provide some isolation from ground planes. Use the standard probes, with the supplied common mode choke (3 turns) in high common mode slew rate situations. It helps to keep the probe cable away from metal surfaces to keep capacitance down. We have verified the probe jacket has a break down voltage of >20 kV.

For particularly fast common mode rise/fall time (<20ns) we can help with a short probe with lower ringing. It is supplied with a short x10 oscilloscope probe fitted with a common mode choke. The aim is to damp the ringing of probe cable inductance and free space capacitance of the IsoPod.



The Probe tip can be fitted with a ground spring or a BNC adaptor. Keeping the ground inductance of the measurement as low as possible is critical for achieving high common mode rejection.



The x10 probe will give a measurement range of +/-8V (0.8V input range) or +/-80V (8V input range). The IsoPod can also be used with other standard oscilloscope probes or coaxial connections.

Heat

The IsoPod generates about 3.5W of heat. To keep the internal temperature down, it is important that the body of the probe has circulating air around it, and the finned surface is upwards, and not covered. The probe will become warm to the touch. If the internal temperature becomes too high, it will shut down, and the indicating leds will change to Red or off if it is too much). Wait for the unit to cool down.